

Osama Mohammed Esmat Hasan

New Cairo, Egypt • +20 127 530 3422 • +20 127 117 2000 • [gmail](#) • [github](#) • [linkedin](#)

Computer Science student specializing in Artificial Intelligence - Future University in Egypt

Professional Summary

Computer Science & AI undergraduate with hands-on experience across computer vision, machine learning, and modern AI systems — including real-time detection models (CNNs, YOLOv8) and Generative AI applications built with LLMs, RAG pipelines, LangChain, and autonomous agents with memory. Skilled in translating theoretical concepts into practical, data-driven solutions.

Education

Future University in Egypt (FUE)

Cairo, Egypt

B.Sc. in Computer Science & Artificial Intelligence — GPA: 3.2 / 4.0

Graduation: June 2027

Relevant Coursework: Machine Learning, Computer Vision, Neural Networks & Deep Learning, Natural Language Processing, Intelligent Agents, Data Structures & Algorithms.

Experience

Digital Egypt Pioneers Initiative (DEPI) — Machine Learning Engineer Track (in partnership with Microsoft) **Cairo, Egypt**
Trainee & Team Lead **July 2026 – December 2026**

Selected for DEPI's Machine Learning Engineer track, an intensive 6-month program in partnership with Microsoft covering applied ML and AI engineering.

Projects

AI Education Platform

FUE, Cairo

Generative AI Project

2026

- Built an AI-powered learning platform featuring quizzes, flashcards, PDF chat, and personalized learning paths using LLMs and RAG.

NutriMind AI

ODC, Giza

Generative AI & RAG Project

2026

- Built an AI-powered InBody analysis and personalized nutrition assistant using RAG, LangChain, memory, and LLMs.

Driver Drowsiness Detection System

FUE, Cairo

Computer Vision & AI Engineering Project

2025

- Built a real-time driver fatigue detection system using custom CNNs and YOLOv8, tuned across 20–30 training epochs.

License Plate Recognition System

FUE, Cairo

Computer Vision & AI Engineering Project

2025

- Built an OpenCV pipeline to detect plate boundaries and extract alphanumeric characters.

Business Intelligence & Software Architecture Implementations

FUE, Cairo

Data Analyst & Software Developer

2025 – 2026

- Built a fake news detection pipeline on a Star Schema data warehouse with Power BI/DAX dashboards.
- Benchmarked Brute Force vs. Dynamic Programming runtime and memory usage for the 0/1 Knapsack problem.

Leadership & Activities

RoboCup Junior Egypt

AASTMT, Alexandria, Egypt

Team Participant

April 2025

- Participated in RoboCup Junior Egypt, a national robotics competition held at the Arab Academy for Science, Technology & Maritime Transport (AASTMT), collaborating with a team on robotics design and problem-solving challenges.
-

Skills & Interests

Programming Languages: Python, Java, SQL, C++

AI / ML & Computer Vision: CNNs, YOLOv8, OpenCV, Machine Learning, Deep Learning

Generative AI: LLMs, RAG, LangChain, AI Agents, Prompt Engineering

Frameworks & Tools: Flask, Streamlit, Pygame, Java Swing (GUI), Power BI (DAX)

Networking & Systems: Cisco IOS, Packet Tracer, Tinkercad

Languages: English (A2)

Interests: Deep Learning Research, Algorithmic Problem Solving, Device & Network Configuration, Robotics Simulation